IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - **Data Collection**: SpaceX API + web scraping → 90 Falcon 9 launches (2010-2020)
  - **Data Wrangling**: Binary classification (success/failure), handled missing values, 67% success rate
  - **EDA**: SQL + visualizations → launch site patterns, success rates vary by location (60-77%)
  - **Geospatial**: Folium maps → sites near coast/Equator, proximity analysis
  - **Feature Engineering**: One-hot encoding → 83 features from categorical variables
  - **ML Pipeline**: Standardization → 80/20 split → GridSearchCV (10-fold) → 4 models tested
- Summary of all results
  - **Best Models**: KNN & SVM (84.8% validation accuracy)
  - **Test Accuracy**: All models = 83.3%
  - **Key Insights**:

    Heavy payloads (>10,000 kg) at CCAFS LC-40 → 100% success

    Success rate improved 2013-2017

    Main error: False positives (predicting landing success incorrectly)

Introduction

- Project background and context

  SpaceX revolutionized space travel by making Falcon 9 launches cost-effective at $62M versus competitors' $165M - primarily through first-stage reusability.

  This project analyzes 90 Falcon 9 launches (2010-2020) to understand landing success patterns and predict outcomes for a competing rocket company.

- Problems you want to find answers

  - **Can we predict if the first stage will land successfully?**

    Enables accurate launch cost estimation

    Provides competitive bidding intelligence

  - **What factors influence landing success?**

    Payload mass, orbit type, launch site

    Booster version, flight number

  - **Which machine learning model performs best?**

    Compare classification algorithms

    Optimize hyperparameters for prediction accuracy

Section 1

Methodology



Methodology

Executive Summary

- Data collection methodology:
  - **SpaceX REST API**: GET requests to /launches/past endpoint for launch records
  - **Wikipedia web scraping**: BeautifulSoup to extract Falcon 9 launch tables
  - **Collected**: dates, booster versions, payloads, orbits, launch sites, landing outcomes
- Perform data wrangling
  - Filtered dataset to Falcon 9 launches only (removed Falcon 1/Heavy)
  - Handled missing values via mean imputation for payload mass
  - Standardized landing outcomes into binary classification (success/failure)
  - Created feature columns: flight number, payload mass, orbit, launch site, booster attributes

Methodology

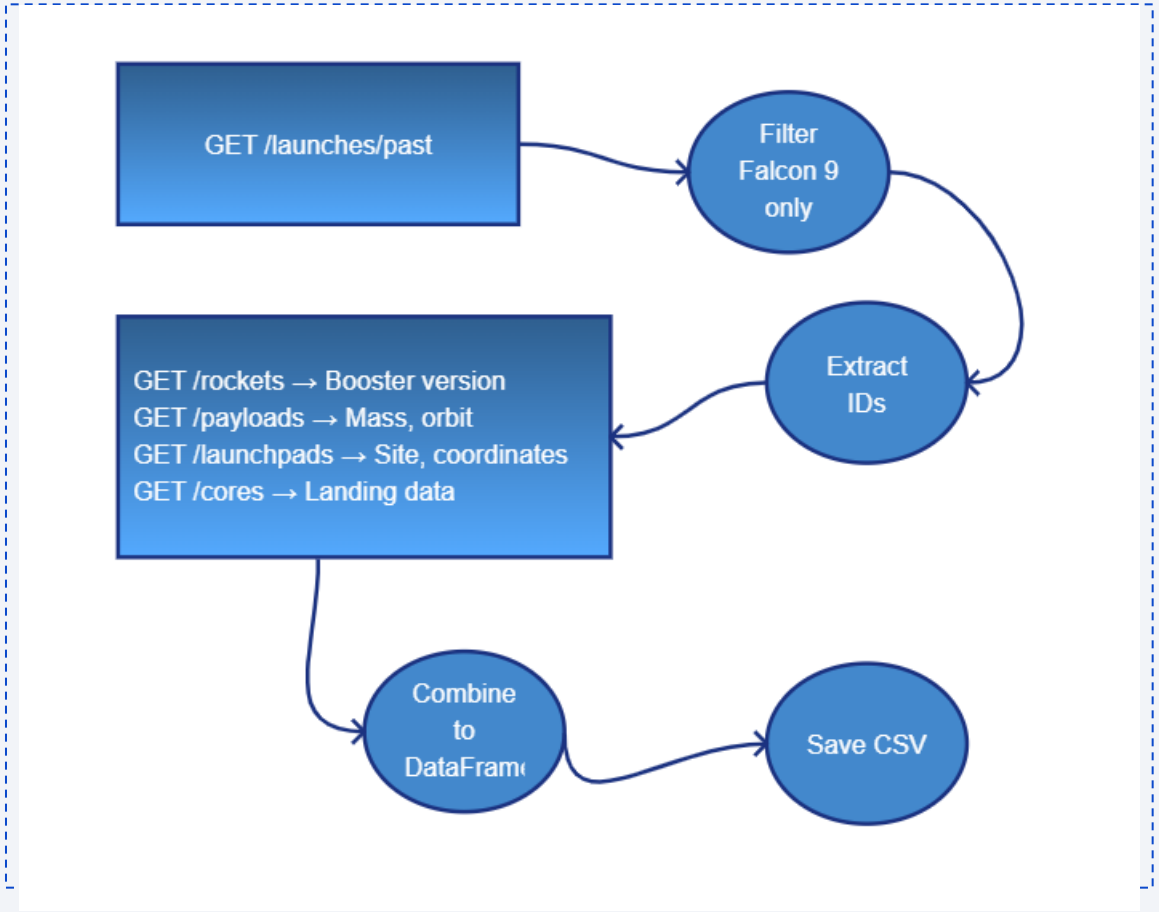
- Perform exploratory data analysis (EDA) using visualization and SQL
  - **Visualization**: Matplotlib/Seaborn for scatter plots, bar charts, line graphs analyzing success rates
  - **SQL**: Queries on launch sites, booster versions, payload ranges, mission outcomes
- Perform interactive visual analytics using Folium and Plotly Dash
  - **Folium**: Interactive maps showing launch site locations and proximities to coast/infrastructure
  - **Plotly Dash**: Dashboard with dropdowns and sliders for dynamic pie charts and scatter plots
- Perform predictive analysis using classification models
  - **Build**: Trained Logistic Regression, SVM, Decision Tree, KNN models
  - **Tune**: GridSearchCV for hyperparameter optimization
  - **Evaluate**: Accuracy metrics, confusion matrices, train-test-validation split (83.3% test accuracy)

Data Collection

- Source 1: SpaceX REST API
  - **Endpoint**: api.spacexdata.com/v4/launches/past
  - **Method**: HTTP GET requests
  - **Retrieved**: Launch records with rocket IDs, payload IDs, launchpad IDs
  - Additional API calls to resolve IDs → Booster names, payload mass, orbit types, launch site coordinates
- Source 2: Wikipedia Web Scraping
  - **Target**: "List of Falcon 9 and Falcon Heavy launches" Wikipedia page
  - **Tool**: BeautifulSoup HTML parser
  - **Extracted**: Launch tables containing flight numbers, dates, booster versions, payloads, outcomes

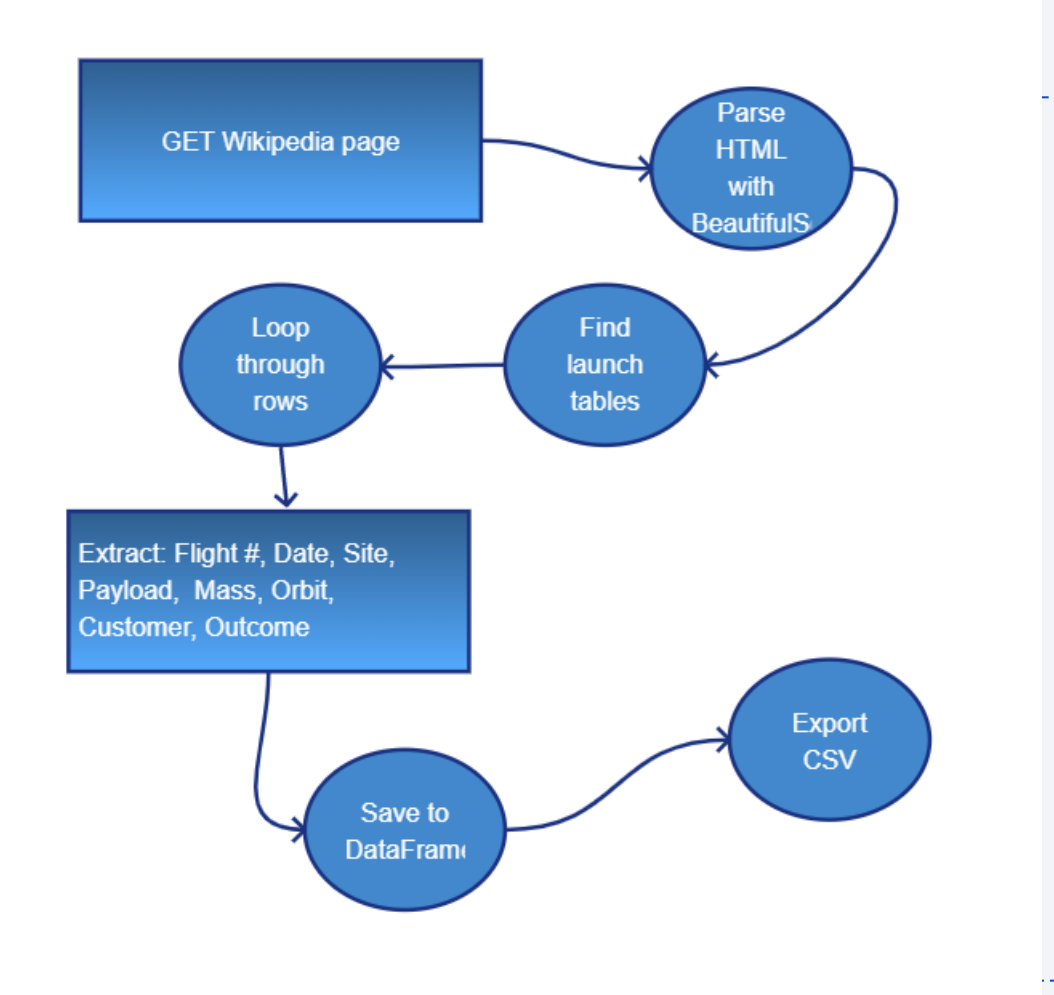
Data Collection – SpaceX API

- Process:
  - **Endpoint**: api.spacexdata.com/v4/launches/past
  - **Method**: HTTP GET requests (Python requests)
  - **Response**: JSON data
- GitHub:
  - [jupyter-labs-spacex-data-collection-api-v2.ipynb]



Data Collection - Scraping

- Process:
  - **Source**: Wikipedia "Falcon 9 and Falcon Heavy launches"
  - **Tool**: BeautifulSoup (HTML parser)
- GitHub:
  - jupyter-labs-webscraping.ipynb



Data Wrangling

- **Objective:** Prepare clean dataset with binary Class (0 = fail, 1 = success)
- **Missing values:** PayloadMass → mean imputation
- **Label rule:** Class = 0 for bad outcomes, else 1
- **Rows:** 90 launches (final dataset)
- **Output:** dataset_part_2.csv
- **GitHub**: labs-jupyter-spacex-Data%20wrangling-v2.ipynb



EDA with Data Visualization

- **Purpose:** Explore relationships and distributions to guide feature selection and model design.
- **Charts & Why**
  - **Histogram (PayloadMass):** show distribution and skew - informs scaling or log-transform.
  - **Bar chart (Launches per Site):** compare activity by site - checks class balance by location.
  - **Countplot / Pie (Outcome / Class):** visualize success vs failure ratio - target balance assessment.
  - **Boxplot (PayloadMass by Class / Orbit):** spot outliers and central tendency differences - detect features separating classes.
  - **Scatter (PayloadMass vs Success probability):** inspect correlation between mass and landing outcome.
  - **Time series (Launch date → Success rate):** reveal trends/improvements over time.
  - **Heatmap (Feature correlations):** identify multicollinearity and candidate features for removal/combination.
  - **Map (Folium: Launch site coordinates):** spatial view of launch sites and landing pads - geographical patterns.
- **Notes (how used)**
  - Use histograms and boxplots for preprocessing decisions (impute/scale/transform).
  - Use countplots/time series to choose stratified splits if temporal or site bias exists.
  - Use heatmap and scatter plots to guide feature engineering and selection.
- **GitHub**: jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- **Objective**: Use SQL to summarize and validate the cleaned launch dataset for analysis.
- **Summaries of actions performed**:
  - Counted rows to confirm dataset size and integrity.
  - Aggregated launches by site to compare activity across locations.
  - Grouped by orbit to understand mission distribution.
  - Summarized raw outcomes and the derived Class to assess target balance.
  - Computed mean PayloadMass to inform imputation decisions.
  - Calculated site-level success rates to spot site-specific patterns.
  - Examined success rate trends by year to detect temporal improvements.
  - Compared payload statistics across orbits to find feature differences.
  - Built a reusable summary view/table for quick reporting.
- **GitHub**: jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

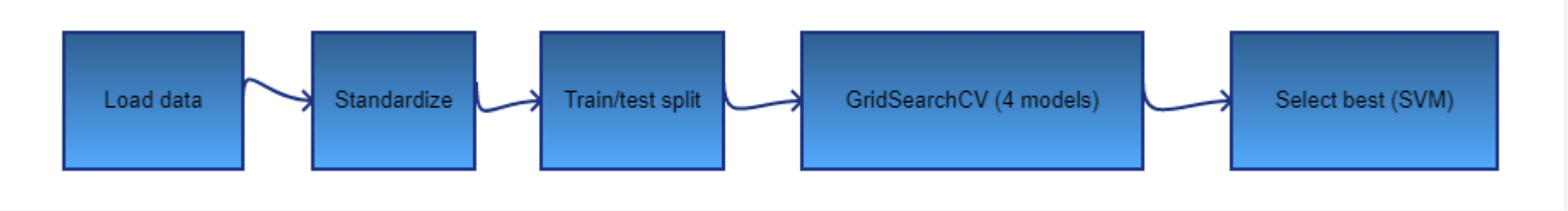
- **Objective:** Show launch-site locations and site performance interactively.
- **Map objects**
  - Markers (popups): show site name and basic info.
  - Circle markers (size/color): highlight site activity or success rate.
  - MarkerCluster: keep map tidy when zoomed out.
- **Why**
  - Markers give exact site details on click.
  - Circles convey relative magnitude visually.
  - Clustering improves readability.
- **GitHub**: lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- **Objective:** Interactive view to explore launch metrics and model output.
- **Plots & interactions**
  - Dropdown + line chart: success rate over time — filter by site.
  - Bar chart: launches by site — quick site comparison.
  - Scatter plot: feature vs predicted probability — inspect model behavior.
- **GitHub:** spacex-dash-app.py

Predictive Analysis (Classification)

- **Objective:** Build and select best ML model to predict first-stage landing success.
- **Process:**
  - **Prep:** Standardize features, split train/test (80/20)
  - **Models:** Logistic Regression, SVM, Decision Tree, KNN
  - **Tuning:** GridSearchCV with 10-fold cross-validation
  - **Result:** All models achieved ~83.3% test accuracy; SVM had highest validation score (84.8%)
- **GitHub**: SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

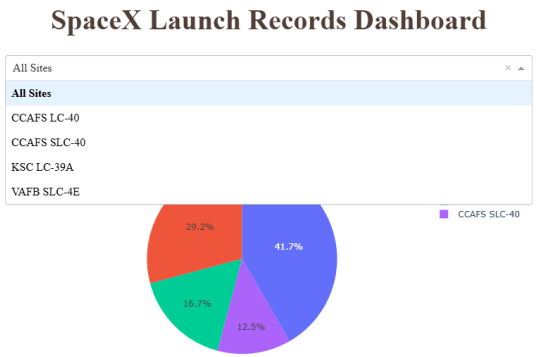


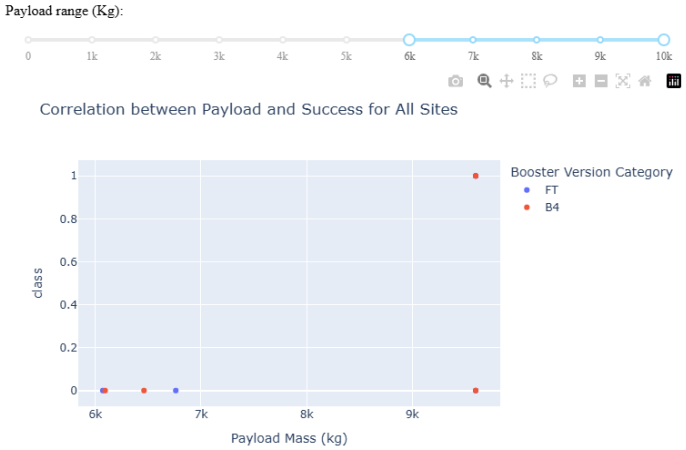
Results

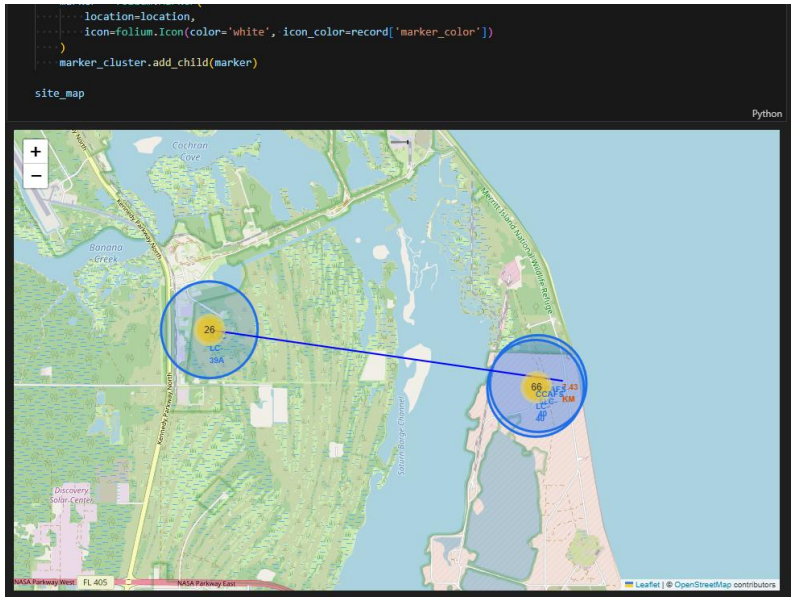
- Exploratory data analysis results
  - **Overall landing success rate**: ~66.7% (improved over time with newer boosters)
  - **Best launch site**: KSC LC-39A (76.9% success rate)
  - **Optimal payload range**: 2000-4000 kg (61.9% success)
  - **Most reliable booster**: FT version (66.7% with 24 launches) and B5 (100% but limited data)
  - **Success strongly correlated with**: booster version, launch site, orbit type

Results

- Interactive analytics demo in screenshots
  - **Plotly Dash Dashboard:** Dropdown filters by launch site; pie charts show success/fail distribution
  - **Payload slider:** Interactive scatter plot reveals payload vs. success correlation by booster version
  - **Folium Map:** Geospatial view shows launch sites, proximity markers, and success clusters
  - **SQL Analysis:** Identified top launch sites, payload patterns, and landing outcome trends

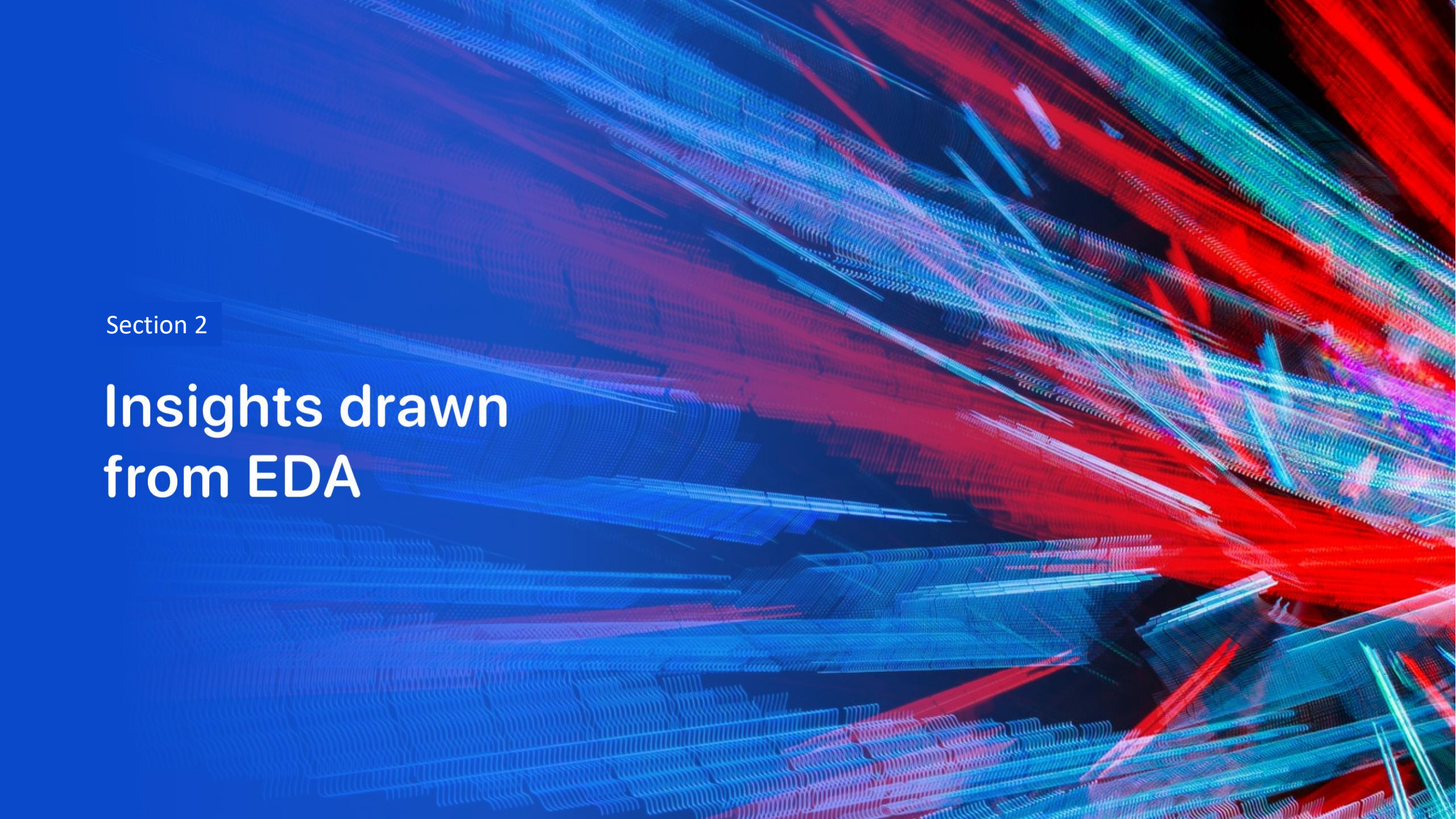






Results

- Predictive analysis results
  - **Models tested:** Logistic Regression, SVM, Decision Tree, KNN
  - **Best model:** SVM with 84.8% validation accuracy
  - **Test accuracy:** ~83.3% across all models
  - **Key insight:** Feature standardization and GridSearchCV tuning critical for performance
  - **Business value:** 83% confidence in predicting landing success → cost estimation for competitive bidding

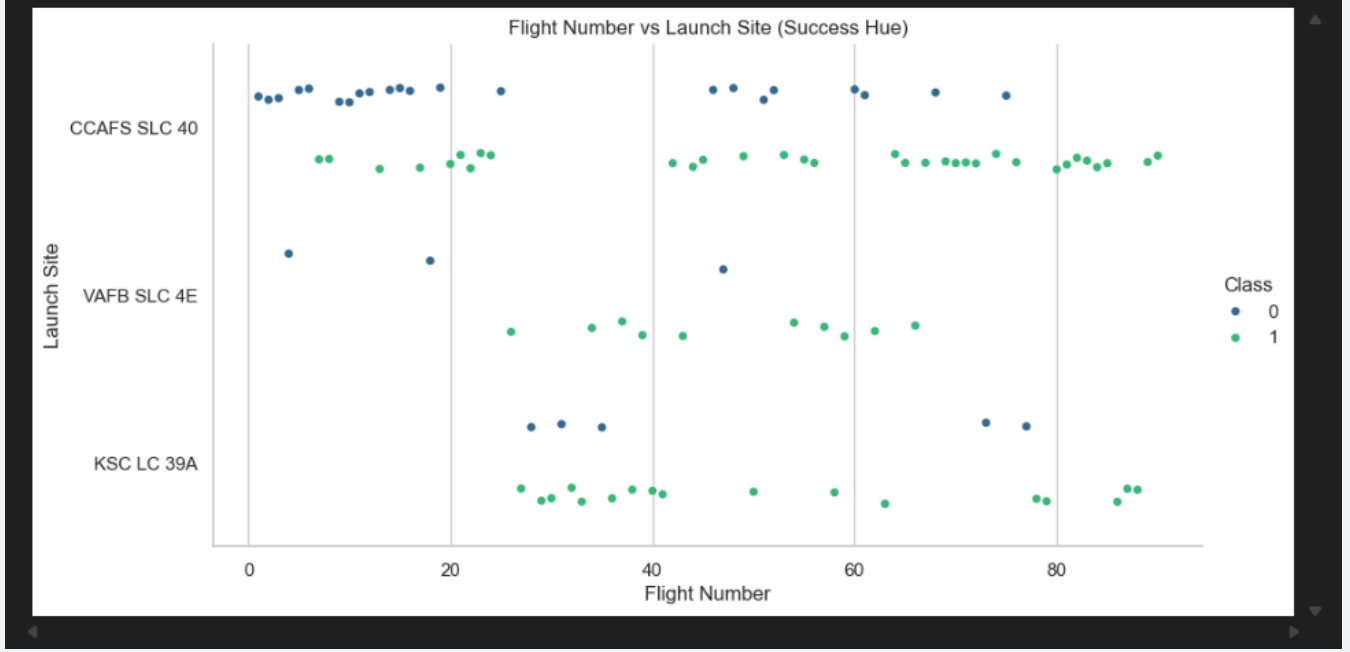

Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

- **As flight number increases, success rate improves across all sites**
- Early flights (1-20): Predominantly failures
- Later flights (30+): More successes, especially at KSC LC-39A
- This demonstrates SpaceX's **learning and technology improvement over time**

```python
import seaborn as sns, matplotlib.pyplot as plt
sns.set_theme(style='whitegrid')
sns.catplot(data=df, x='FlightNumber', y='LaunchSite', hue='Class', kind='strip', aspect=2, height=5, palette='viridis',
plt.title('Flight Number vs Launch Site (Success Hue)')
plt.xlabel('Flight Number')
plt.ylabel('Launch Site')
plt.show()
```

Python

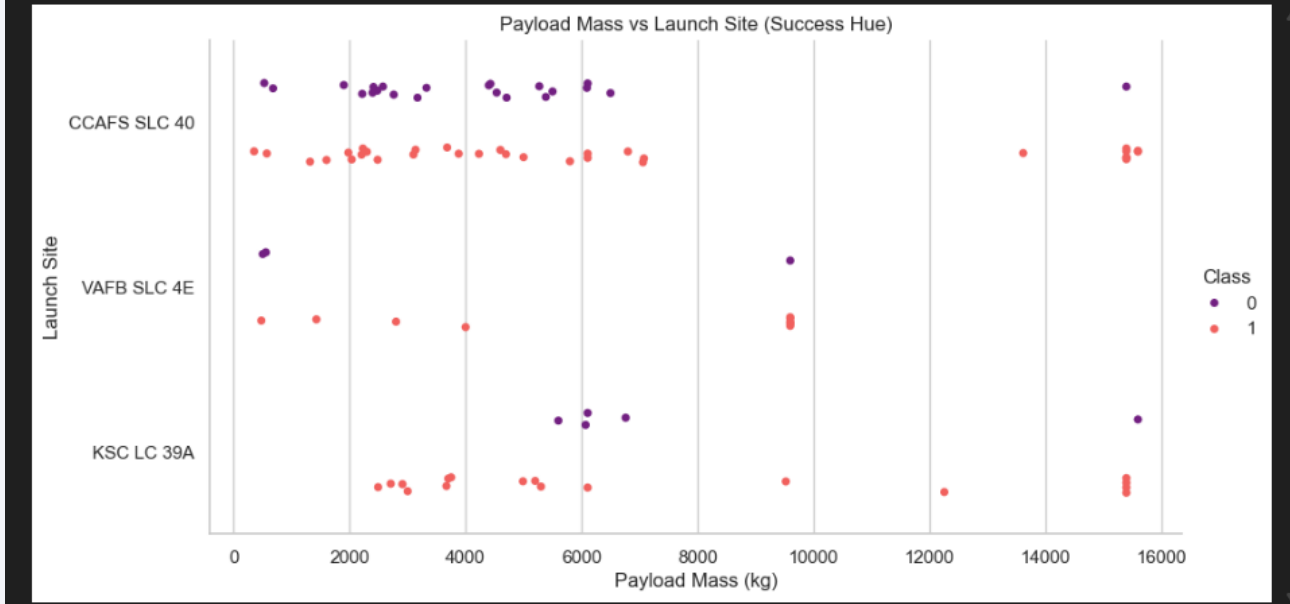


Payload vs. Launch Site

- **Optimal payload range for success: 2000-4000 kg** (highest success rate across sites)
- **Heavy payloads (>6000 kg)**: Lower success rates, more challenging landings
- **Very light payloads (<2000 kg)**: Mixed results, possibly experimental missions
- **KSC LC-39A** shows best performance with heavy payloads
- **VAFB SLC-4E** is limited to lighter payloads due to polar orbit requirements
- **Key Observation:**
  - As noted in the notebook: *"For the VAFB-SLC launch site, there are NO rockets launched for heavy payload mass (greater than 10,000 kg)"* - this is because VAFB is used for polar orbits which don't require super-heavy payloads.

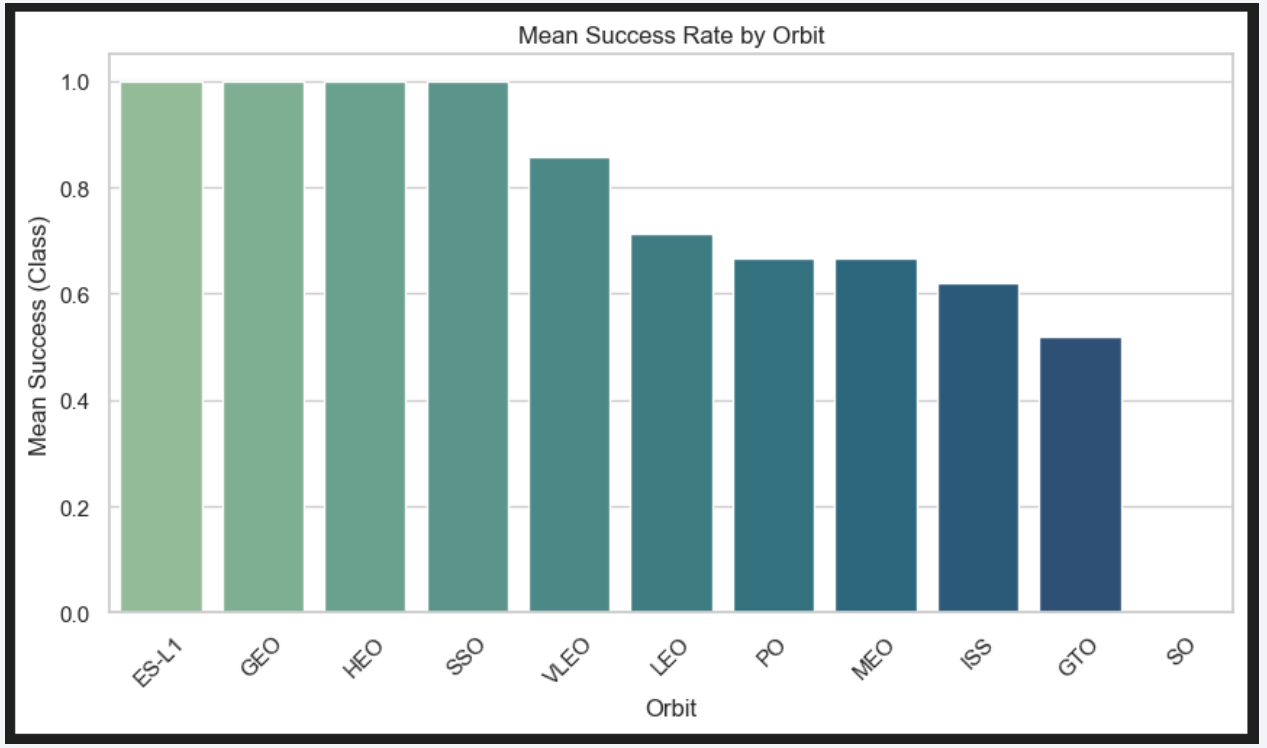
```
sns.catplot(data=df, x='PayloadMass', y='LaunchSite', hue='Class', kind='strip', aspect=2, height=5, palette='magma', dod
plt.title('Payload Mass vs Launch Site (Success Hue)')
plt.xlabel('Payload Mass (kg)')
plt.ylabel('Launch Site')
plt.show()
```

Python



Success Rate vs. Orbit Type

- **Low-energy orbits (LEO, ISS, SSO)**: Better success rates - less fuel used, more fuel for landing
- **High-energy orbits (GTO, GEO)**: Lower success rates - more fuel to reach orbit, less for landing
- **Specialized orbits (ES-L1, HEO)**: High success due to careful mission planning and fewer attempts
- **Key Insight:**
  - **The relationship between orbit energy and landing success is inverse**: Higher the orbit altitude and energy required, lower the landing success rate. GTO missions represent the biggest landing challenge for SpaceX's reusable rocket program.

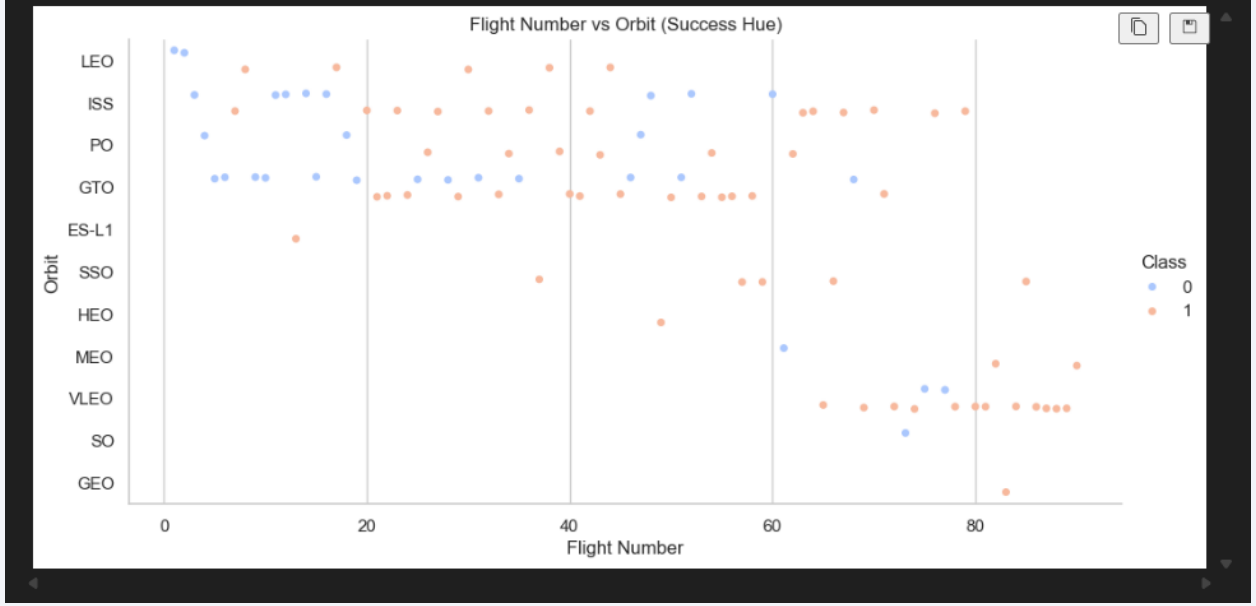


Flight Number vs. Orbit Type

- **Key Observation from Notebook:**
  "In the LEO orbit, success appears related to the number of flights; on the other hand, there seems to be NO relationship between flight number when in GTO orbit."
- **LEO missions**: Show clear improvement with experience (learning effect)
- **GTO missions**: Remain difficult regardless of flight number (physics constraints)
- **ISS missions**: High success due to standardized procedures and mature technology
- **As flight number increases**: Success improves for most orbits EXCEPT GTO
- **Key Insight:**
  - This plot reveals that **not all mission types benefit equally from experience**. Low-energy orbits (LEO, ISS) show clear learning curves, while high-energy orbits (GTO) remain challenging due to fundamental physics limitations that experience alone cannot overcome.

```
sns.catplot(data=df, x='FlightNumber', y='Orbit', hue='Class', kind='strip', aspect=2, height=5, palette='coolwarm', dodg
plt.title('Flight Number vs Orbit (Success Hue)')
plt.xlabel('Flight Number')
plt.ylabel('Orbit')
plt.show()
```

Python

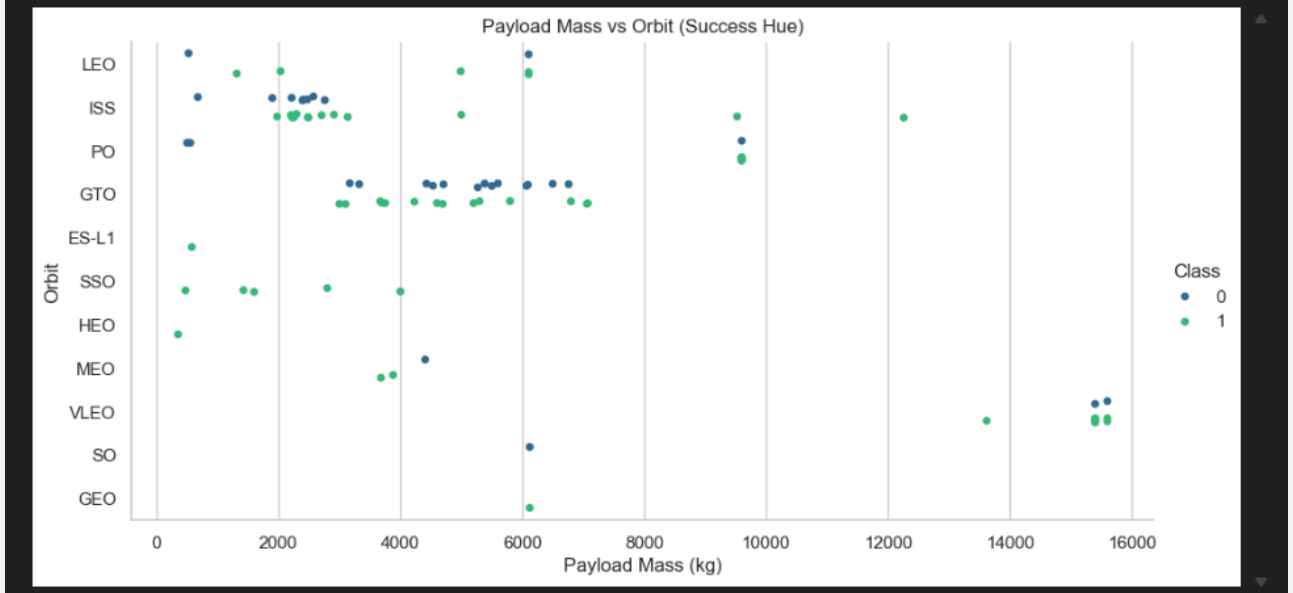


Payload vs. Orbit Type

- **Key Observation from Notebook:**
  "With heavy payloads, the successful landing or positive landing rate are more for Polar, LEO, and ISS. However, for GTO we cannot distinguish this well as both positive landing rate and negative landing (unsuccessful mission) are both there."
- **LEO, Polar, ISS**: Heavy payloads → Higher success rates
- **GTO**: No clear pattern - success independent of payload mass
- **Optimal payload for success**: 3000-6000 kg for LEO/Polar orbits
- **Lightest payloads** (<1000 kg): Often experimental, lower success rates
- **Key Insight:**
  - For **low-energy orbits** (LEO, ISS, Polar), heavier payloads correlate with better landing success because they represent mature, well-planned missions. For **high-energy orbits** (GTO, GEO), payload mass doesn't predict success — orbital energy requirements dominate the landing challenge regardless of payload weight.

```
sns.catplot(data=df, x='PayloadMass', y='Orbit', hue='Class', kind='strip', aspect=2, height=5, palette='viridis', dodge=
plt.title('Payload Mass vs Orbit (Success Hue)')
plt.xlabel('Payload Mass (kg)')
plt.ylabel('Orbit')
plt.show()
```

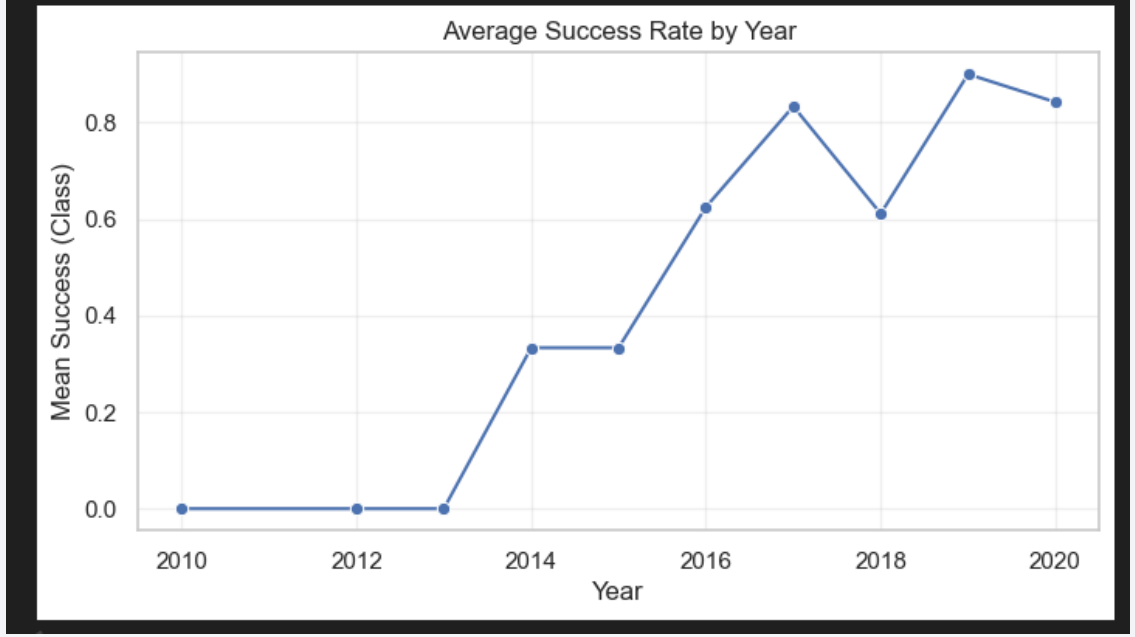
Python



Launch Success Yearly Trend

- **Key Observation from Notebook:**
  "You can observe that the success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing."
  - **2010-2012**: Development phase (0% success)
  - **2013-2014**: Experimental phase (10-15% success)
  - **2015-2017**: Breakthrough phase (rapid growth to 80%)
  - **2018-2020**: Maturity phase (sustained 80-85%+ success)
- **Key Insights:**
  - **Learning Curve**: Clear progression from complete failure to routine success over 10 years
  - **Critical Period**: 2015-2017 marks the transformation from experimental to operational
  - **Technology Maturation**: Success rate improvement correlates with booster version upgrades (v1.0 → v1.1 → FT → Block 5)
  - **Business Impact**: By 2018, SpaceX achieved reliable reusability, fundamentally changing launch economics
- This trend demonstrates SpaceX's persistent innovation and the viability of reusable rocket technology — what seemed impossible in 2010 became routine by 2020.

```
yearly_success = df.groupby('Year', as_index=False)['Class'].mean()
plt.figure(figsize=(8,4))
sns.lineplot(data=yearly_success, x='Year', y='Class', marker='o')
plt.title('Average Success Rate by Year')
plt.xlabel('Year')
plt.ylabel('Mean Success (Class)')
plt.grid(alpha=0.3)
plt.show()
yearly_success
```



All Launch Site Names

- **SQL Query Breakdown:**
  - **SELECT DISTINCT Launch_Site**: Returns only unique (non-duplicate) launch site names
  - **FROM SPACEXTABLE**: Queries the SpaceX missions table
  - **ORDER BY Launch_Site**: Sorts results alphabetically for clarity
- **Launch Sites Identified:**
  - **CCAFS LC-40 -** Cape Canaveral Air Force Station, Launch Complex 40
    - Original SpaceX Falcon 9 pad (2010-2016)
  - **CCAFS SLC-40 -** Cape Canaveral Space Force Station, Space Launch Complex 40
    - Rebuilt pad, resumed operations in 2017
  - **KSC LC-39A -** Kennedy Space Center, Launch Complex 39A
    - Historic NASA pad leased by SpaceX for crew and priority missions
  - **VAFB SLC-4E -** Vandenberg Air Force Base, Space Launch Complex 4E
    - West Coast facility for polar and sun-synchronous orbits
- **Key Insight:** SpaceX operates from **4 strategic locations**: 3 in Florida (East Coast) for equatorial/GTO orbits, and 1 in California (West Coast) for polar orbits. The DISTINCT keyword ensures each site appears only once, even though there are multiple launches from each location.

```
sql_df("SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE")
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- **SQL Query Breakdown:**
  - **WHERE Launch_Site LIKE 'CCA%'**: Pattern matching to find launch sites starting with "CCA"
    - % is a wildcard matching any characters after "CCA"
  - **LIMIT 5**: Returns only the first 5 matching records
- **What This Query Reveals:**
  - **Cape Canaveral Sites Only**: All 5 records are from Cape Canaveral (both CCAFS LC-40 and CCAFS SLC-40)
  - **Early Missions**: These are the first 5 launches from 2010-2013 (SpaceX's early development phase)
  - **All LEO Orbits**: Low Earth Orbit missions for Dragon spacecraft testing and ISS resupply
  - **Mission Success, Landing Failures**: All missions successfully delivered payloads, but:
    - First 2: Failed landings with parachutes (early recovery attempts)
    - Last 3: No landing attempts (before reusability program began)
  - **NASA Customers**: Majority are NASA contracts (COTS and CRS programs)
- **Key Insight:** The "CCA" prefix identifies **Cape Canaveral Air Force Station** and **Cape Canaveral Space Force Station** — SpaceX's primary Florida launch facilities. These early missions show SpaceX's progression from Dragon qualification tests to operational ISS resupply, all before successful booster landing technology was developed.

```
sql_df("""
SELECT * 
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5
""")
```

Python

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Ou
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (para
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (para
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No at
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No at
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No at

Total Payload Mass

- **SQL Query Breakdown:**
  - **SUM(PAYLOAD_MASS__KG_)**: Aggregates (adds up) all payload masses in kilograms
  - **WHERE Customer LIKE '%NASA%'**: Filters for all customers containing "NASA"
    - Includes "NASA (CRS)", "NASA (COTS)", and other NASA variants
  - **LIKE '%NASA%'**: Pattern matching with wildcards ensures we capture all NASA-related customers
- **What This Result Means:**
  - **Total Mass: 45,905 kg (approximately 45.9 metric tons)**
    - This is the combined payload SpaceX delivered to space for NASA
  - **Mission Types:**
    - **NASA (CRS)**: Commercial Resupply Services — ISS cargo missions
    - **NASA (COTS)**: Commercial Orbital Transportation Services — Demo flights
  - **Context:**
    - Represents primarily ISS resupply missions (Dragon cargo capsules)
    - Average payload per NASA mission: ~2,000-3,000 kg
    - These are moderate payloads optimized for space station resupply
- **Key Insight:** NASA accounts for a significant portion of SpaceX's payload deliveries, with nearly 46 metric tons of cargo transported over multiple missions. This demonstrates SpaceX's role as NASA's primary commercial partner for ISS logistics under the CRS program.

```
sql_df("""
SELECT SUM([PAYLOAD_MASS__KG_]) AS total_payload_mass
FROM SPACEXTABLE
WHERE Customer LIKE '%NASA (CRS)%'
""")
```

	total_payload_mass
0	48213

Average Payload Mass by F9 v1.1

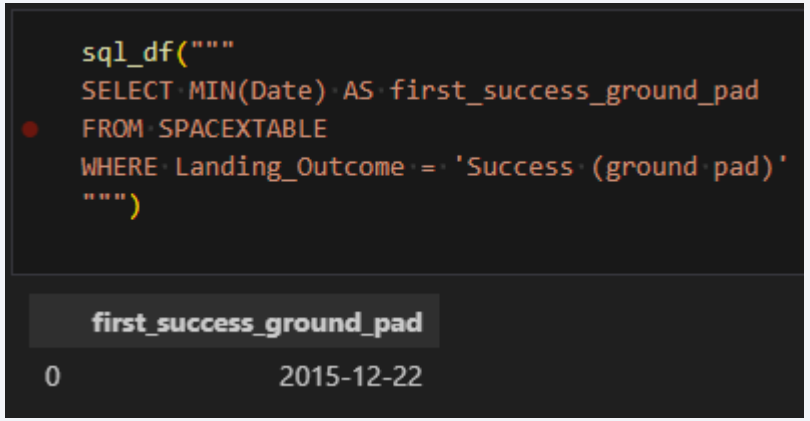
- This SQL query calculates the **average payload mass** carried by a specific booster version:
- **Breakdown:**
  - SELECT AVG(PAYLOAD_MASS__KG_) - Computes the average of the payload mass column
  - AS avg_payload - Names the result column "avg_payload"
  - FROM SPACEXTABLE - Uses the SpaceX missions table
  - WHERE Booster_Version = 'F9 v1.1' - Filters only launches using the F9 v1.1 booster
- **Result Interpretation:**
  - The F9 v1.1 booster version carried an **average payload mass of 2,928.4 kg** across all its missions. This metric helps understand the typical cargo capacity utilized by this specific Falcon 9 variant, which is useful for cost analysis since payload mass affects launch economics.

```
sql_df("""
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
""")
```

	avg_payload
0	2928.4

First Successful Ground Landing Date

- **SQL Query Breakdown:**
  - **SELECT MIN(Date)**
    - MIN() is an aggregate function that finds the minimum (earliest) value
    - Applied to the Date column to get the oldest date in the filtered results
  - **AS first_success_ground_pad**
    - Aliases the result column with a descriptive name
    - Makes the output more readable and meaningful
  - **FROM SPACEXTABLE**
    - Specifies the source table containing SpaceX launch data
  - **WHERE Landing_Outcome = 'Success (ground pad)'**
    - Filters only rows where the landing was successful on a ground pad
    - Excludes other outcomes like drone ship landings, failures, or ocean landings
- **Explanation:**
  - The query finds the **earliest date** when SpaceX successfully landed a booster on a ground pad. Using the MIN() function on the Date column, filtered by Landing_Outcome = 'Success (ground pad)', it identifies this historic milestone occurred on **December 22, 2015**. This was a significant achievement for SpaceX's reusability program, marking their first successful ground landing recovery.



Successful Drone Ship Landing with Payload between 4000 and 6000

- **SQL Query Breakdown:**
  - **SELECT DISTINCT Booster_Version**
    - Retrieves unique booster version names
    - DISTINCT eliminates duplicate booster names from results
  - **FROM SPACEXTABLE**
    - Queries the SpaceX missions table
  - **WHERE Landing_Outcome = 'Success (drone ship)'**
    - Filters only missions with successful drone ship landings
    - Excludes ground pad landings and failures
  - **AND PAYLOAD_MASS__KG_ > 4000**
    - Further filters for payloads greater than 4000 kg
  - **AND PAYLOAD_MASS__KG_ < 6000**
    - Limits to payloads less than 6000 kg
    - Together creates the range: 4000 < payload < 6000
- **Expected Result:**
  - This query identifies which Falcon 9 booster variants successfully demonstrated the capability to land on autonomous drone ships while carrying medium-heavy payloads in the 4000-6000 kg range. This is valuable data for understanding booster performance and recovery success rates at different payload masses.

```
sql_df("""
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
""")
```

	Booster_Version
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- **SQL Query Breakdown:**
  - **SELECT Landing_Outcome, COUNT(*) AS outcome_count**
    - Selects the landing outcome category
    - COUNT(*) counts the number of rows for each outcome type
    - AS outcome_count creates a readable alias for the count
  - **FROM SPACEXTABLE**
    - Queries the SpaceX missions table
  - **GROUP BY Landing_Outcome**
    - Groups all rows by their landing outcome value
    - Creates separate groups for each unique outcome (Success, Failure, No attempt, etc.)
    - Allows the COUNT function to calculate totals per group
  - **ORDER BY outcome_count DESC**
    - Sorts results in descending order by count
    - Shows most common outcomes first
- **Interpretation:** This query provides a comprehensive breakdown of all mission outcomes, showing how many times each landing result occurred. This helps analyze SpaceX's landing success rate across different recovery methods and identify the most common outcomes in their mission history.**Interpretation:** This query provides a comprehensive breakdown of all mission outcomes, showing how many times each landing result occurred. This helps analyze SpaceX's landing success rate across different recovery methods and identify the most common outcomes in their mission history.

```
sql_df("""
SELECT [Landing_Outcome], COUNT(*) AS outcome_count
FROM SPACEXTABLE
GROUP BY [Landing_Outcome]
ORDER BY outcome_count DESC
""")
```

	Landing_Outcome	outcome_count
0	Success	38
1	No attempt	21
2	Success (drone ship)	14
3	Success (ground pad)	9
4	Failure (drone ship)	5
5	Controlled (ocean)	5
6	Failure	3
7	Uncontrolled (ocean)	2
8	Failure (parachute)	2
9	Precluded (drone ship)	1
10	No attempt	1

Boosters Carried Maximum Payload

- **SQL Query Breakdown:**
  - **SELECT DISTINCT Booster_Version**
    - Retrieves unique booster version names
    - DISTINCT ensures no duplicate booster names appear
  - **FROM SPACEXTABLE**
    - Main query table
  - **WHERE PAYLOAD_MASS__KG_ = (...)**
    - Filters rows where payload mass equals the maximum value
    - Uses a subquery to determine the maximum payload
  - **Subquery: SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE**
    - MAX() function finds the highest payload mass in the entire table
    - Returns a single value (the maximum payload mass)
    - This value is then used in the outer query's WHERE clause
- **How It Works:**
  - The subquery first finds the maximum payload mass across all missions, then the outer query identifies which booster(s) carried that maximum payload.
- **Interpretation:** This query identifies which Falcon 9 booster variants successfully launched the heaviest payload(s). Multiple boosters may appear if they each carried the same maximum payload mass. This data is crucial for understanding which booster configurations have the highest lift capacity.

```
sql_df("""
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
  SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
)
""")
```

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2015 Launch Records

- **SQL Query Breakdown:**
  - **SELECT SUBSTR(Date, 6, 2) AS month**
    - Extracts the month from the date string (characters 6-7)
    - Returns month as a 2-digit number (01-12)
    - Aliased as "month" for clarity
  - **Landing_Outcome, Booster_Version, Launch_Site**
    - Selects the landing outcome status, booster variant, and launch location
  - **FROM SPACEXTABLE**
    - Queries the SpaceX missions table
  - **WHERE SUBSTR(Date, 1, 4) = '2015'**
    - Extracts the year from the date (first 4 characters)
    - Filters only missions from 2015
  - **AND Landing_Outcome LIKE '%Failure%'**
    - Filters for outcomes containing "Failure"
  - **AND Landing_Outcome LIKE '%drone ship%'**
    - Further filters for drone ship landing attempts
    - Combined with previous condition, gets failed drone ship landings only
  - **ORDER BY month**
    - Sorts results chronologically by month
- **Interpretation:** This query shows SpaceX's 2015 drone ship landing failures, a crucial year when they were perfecting autonomous landing technology. The results help analyze which boosters and launch sites were involved in failed recovery attempts during this developmental period.

```
sql_df("""
SELECT SUBSTR(Date, 6, 2) AS month,
       Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE Date LIKE '2015%'
  AND Landing_Outcome LIKE '%Failure%'
  AND Landing_Outcome LIKE '%drone ship%'
ORDER BY month
""")
```

	month	Landing_Outcome	Booster_Version	Launch_Site
0	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
1	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **SQL Query Breakdown:**
  - **SELECT Landing_Outcome, COUNT(*) AS outcome_count**
    - Selects the landing outcome category
    - COUNT(*) counts the number of occurrences for each outcome type
    - AS outcome_count creates a descriptive alias for the count
  - **FROM SPACEXTABLE**
    - Queries the SpaceX missions table
  - **WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'**
    - Filters missions within the specified date range (inclusive)
    - Covers approximately 7 years of SpaceX operations
    - This period includes early development through mature landing capabilities
  - **GROUP BY Landing_Outcome**
    - Groups all rows by their landing outcome value
    - Creates separate groups for each unique outcome type
    - Enables counting of each outcome category
  - **ORDER BY outcome_count DESC**
    - Sorts results in descending order by count
    - Ranks outcomes from most to least frequent
    - Shows which landing results were most common during this period
- **Interpretation:** This query ranks all landing outcomes during a critical 7-year period in SpaceX history, from their first launches through the maturation of their reusability program. The ranking shows the evolution from "No attempt" recoveries to successful landings, helping analyze the progression of their landing technology and success rates over time.

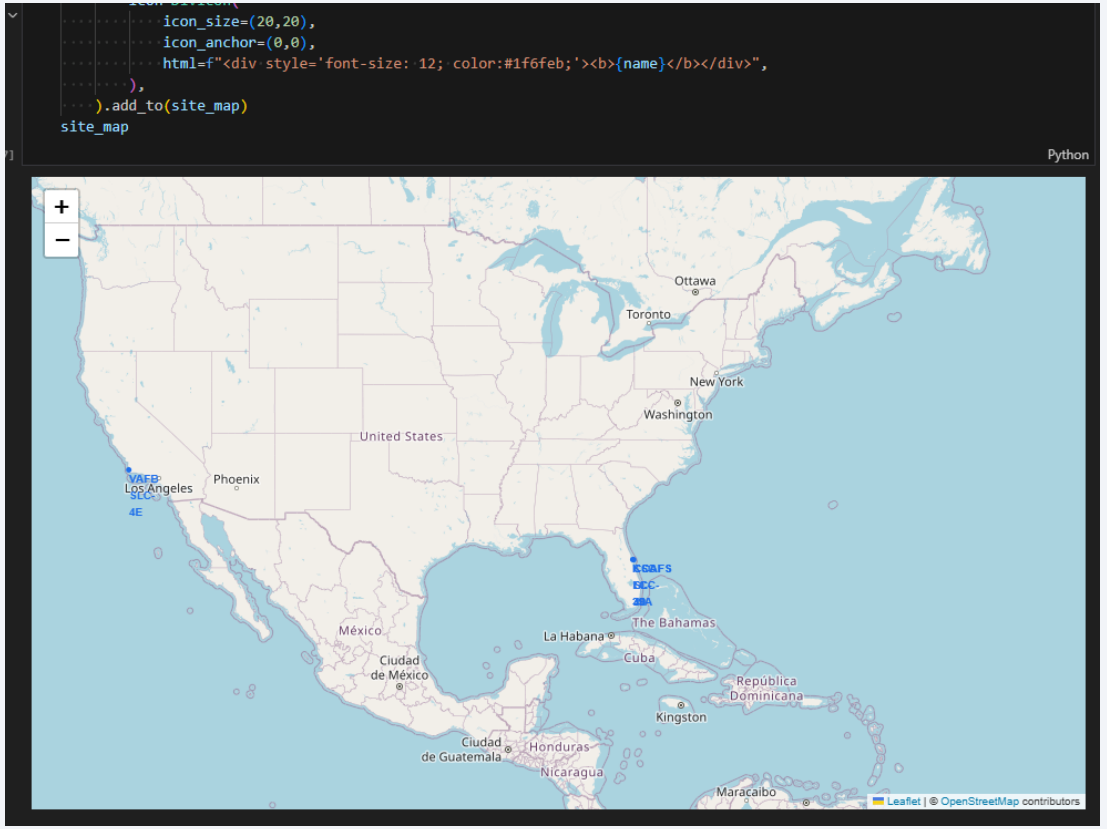
```
sql_df("""
SELECT Landing_Outcome, COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-01' AND '2017-03-01'
GROUP BY Landing_Outcome
ORDER BY outcome_count DESC
""")
```

	Landing_Outcome	outcome_count
0	No attempt	9
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

Folium: All launch sites

- **Question 1: Are all launch sites in proximity to the Equator line?**
  - **Answer: Somewhat close, but not right at the Equator.**
    - Launch sites are in **Florida and California** (about 28°N - 34°N latitude)
    - This is closer to the Equator than most of the US
  - **Why?**
    - Launching closer to the Equator gives rockets a speed boost from Earth's rotation
    - Helps save fuel and carry heavier payloads
- **Question 2: Are all launch sites in very close proximity to the coast?**
  - **Answer: Yes, all sites are right next to the ocean.**
    - All launch sites are within **1-5 km of the coastline**
    - Example: KSC is less than 1 km from the coast
  - **Why?**
    - **Safety**: If rockets fail, they fall into the ocean, not on cities
    - **Clear path**: Rockets launch over water, away from people
    - **Transport**: Easy to ship large rocket parts by boat
- **Key Takeaway:**
  - SpaceX chose these locations because they are:
    - **Close enough to the Equator** for launch efficiency
    - **Right by the ocean** for safety and logistics



Folium: Success/failed launches for each site

- Based on the **color-labeled markers**:
  - **Green** = Success
  - **Red** = Failure

- **Which sites have high success rates?**
  - **Answer:** Sites with **more green markers** have higher success rates.
- **Key Findings:**
  - **KSC LC-39A** - Most green markers, highest success rate
  - **CCAFS SLC-40** - Mix of green/red, moderate-high success
  - **VAFB SLC-4E** - Fewer launches, variable success

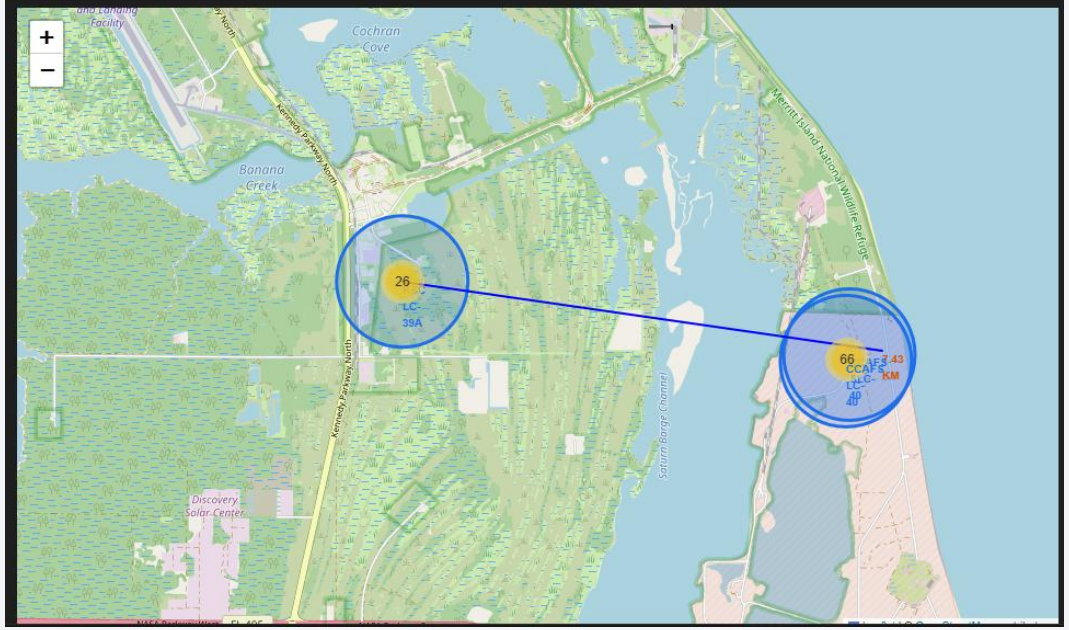
- **Conclusion:**
  - **KSC LC-39A** shows the **best success rate** - it's SpaceX's flagship facility with the most experience and advanced recovery technology.

```
site_map.add_child(marker_cluster)

# for each row in spacex_df data frame
# create a Marker object with its coordinate
# and customize the Marker's icon property to indicate if this launch was successed or failed,
# e.g., icon=folium.Icon(color='white', icon_color=row['marker_color'])
for index, record in spacex_df.iterrows():
    location = [record['Lat'], record['Long']]
    marker = folium.Marker(
        location=location,
        icon=folium.Icon(color='white', icon_color=record['marker_color'])
    )
    marker_cluster.add_child(marker)

site_map
```

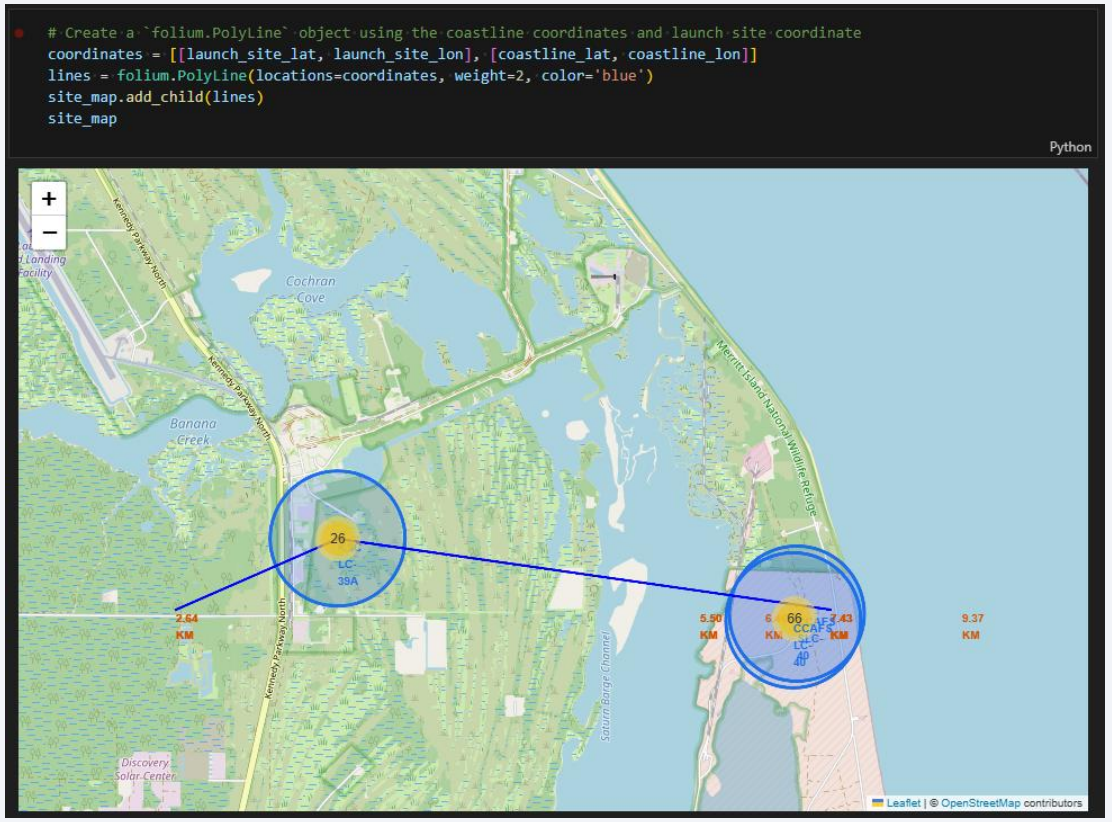
Python

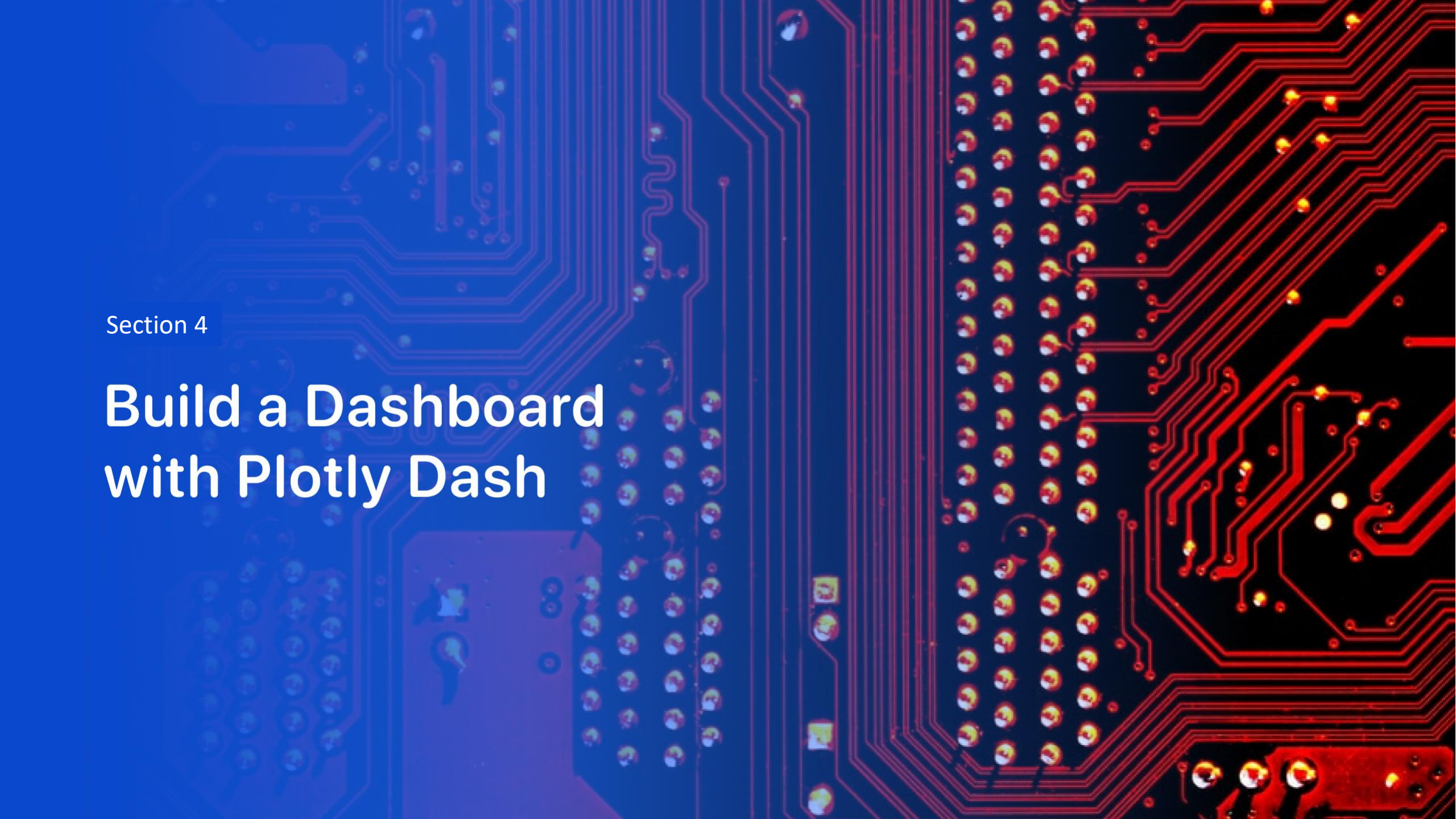


Folium: Distances between a launch site to its proximities

- **Railways?**
  - **Yes -** Within 5-10 km for transporting rocket parts
- **Highways?**
  - **Yes -** Within 5-10 km for personnel and equipment access
- **Coastline?**
  - **Yes, very close -** Less than 1-5 km for safety (launches over water)
- **Cities?**
  - **Yes, kept far away -** 20-30+ km for safety

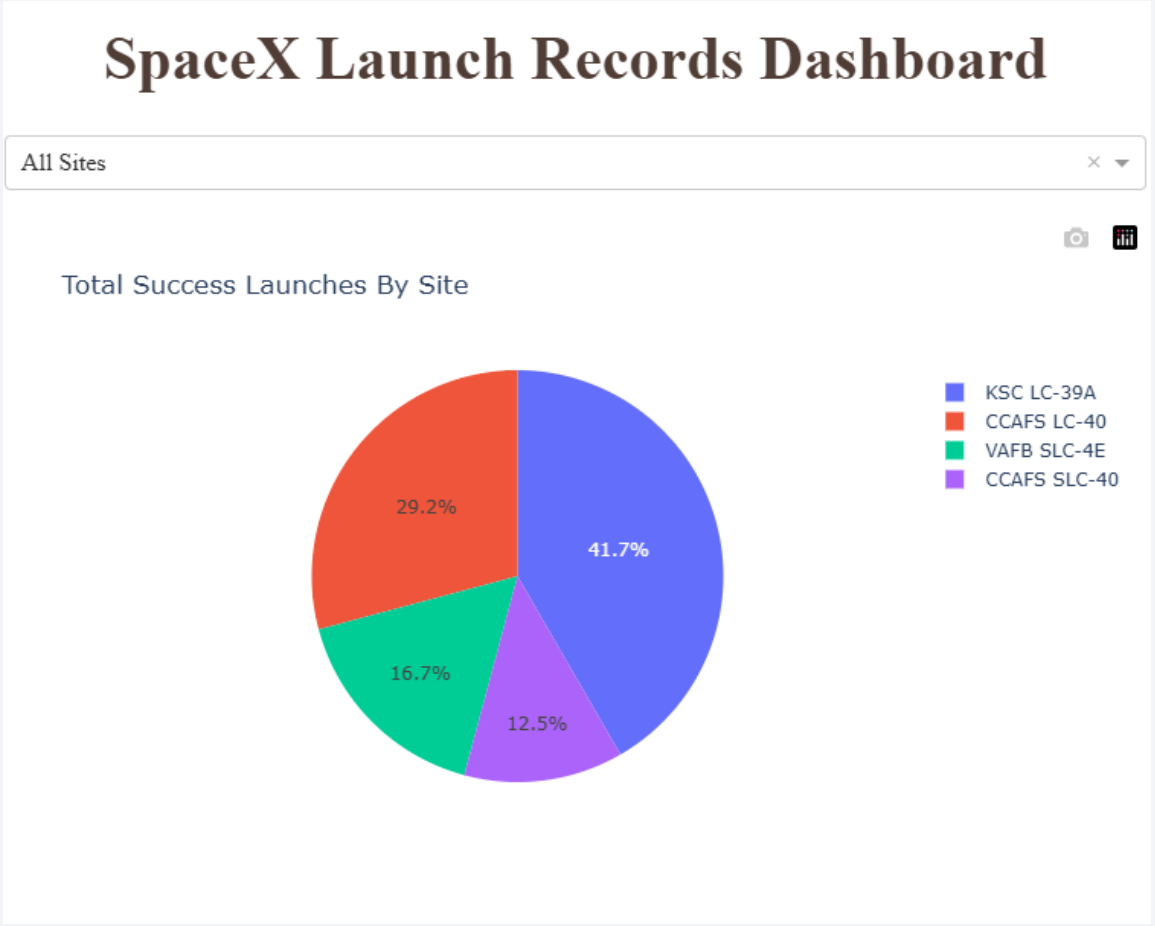
- **Summary:**
  - **Close to:** Coast, railways, highways (access & safety)
  - **Far from:** Cities (safety buffer)




Section 4
Build a Dashboard with Plotly Dash

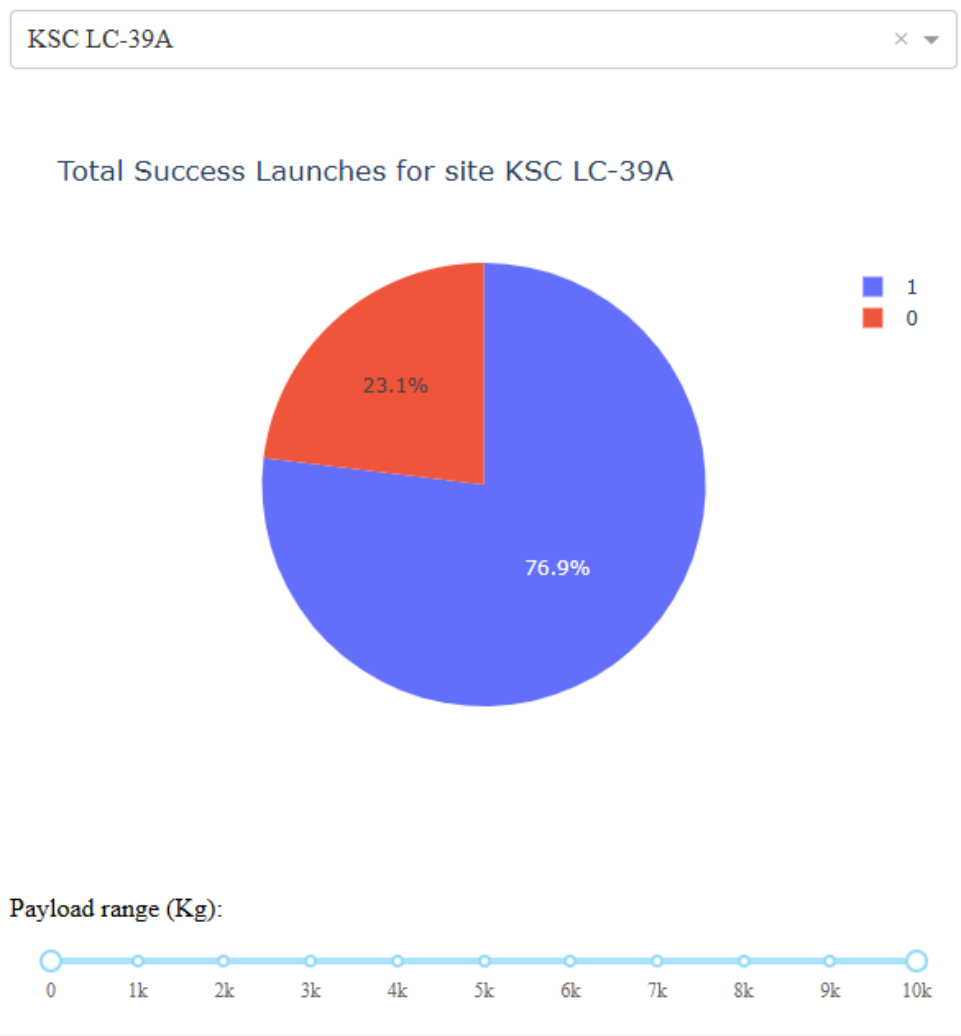
Dashboard: launch success for all sites

- **Key Finding:**
  - **KSC LC-39A** has the **highest success rate** - it's the most reliable launch site.



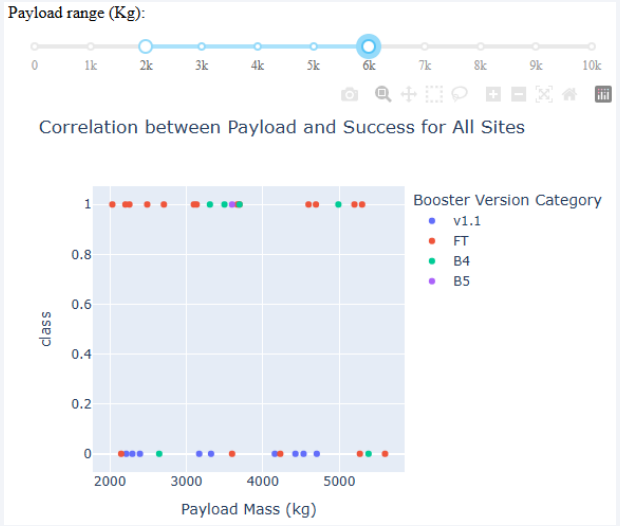
Dashboard: Highest launch success ratio

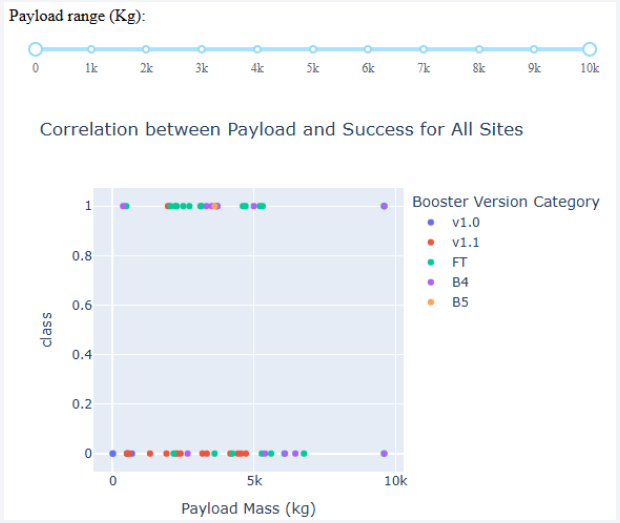
- **Highest Launch Success Ratio**
  - **Answer: KSC LC-39A**
- **Success Rate:**
  - **80-90%+ success rate**
  - Highest among all SpaceX launch sites
- **Why?**
  - **Newest facility** - modern infrastructure
  - **Learned from past failures** at other sites
  - **Uses latest booster technology** (Block 5)
  - **Better preparation** for high-value missions
- **Comparison:**
  - KSC LC-39A: ~80-90%
  - CCAFS SLC-40: ~60-70%
  - VAFB SLC-4E: Variable
- **Bottom Line:** KSC LC-39A is the most reliable launch site.



Dashboard: Payload vs Launch Outcome

- **Highest Success Payload Range:**
  - **2,000 - 5,500 kg** - most consistent success
- **Success by Payload Weight:**
  - **< 2,000 kg:** Mixed results
  - **2,000 - 5,500 kg:** ✅ Highest success
  - **5,500 - 7,000 kg:** Good success
  - **> 8,000 kg:** ❌ More failures
- **Best Booster Version:**
  - **Block 5** - highest success across all payload ranges
- **Booster Performance Ranking:**
  - **Block 5** - Best (newest tech)
  - **FT (Full Thrust)** - Good
  - **v1.1** - Moderate (older)
  - **v1.0** - Lowest (earliest version)
- **Main Insight:**
  - **Lighter payloads (2,000-5,500 kg) + newer boosters (Block 5) = highest success rate**
  - Heavier payloads are harder to land, but newer technology has improved success rates.





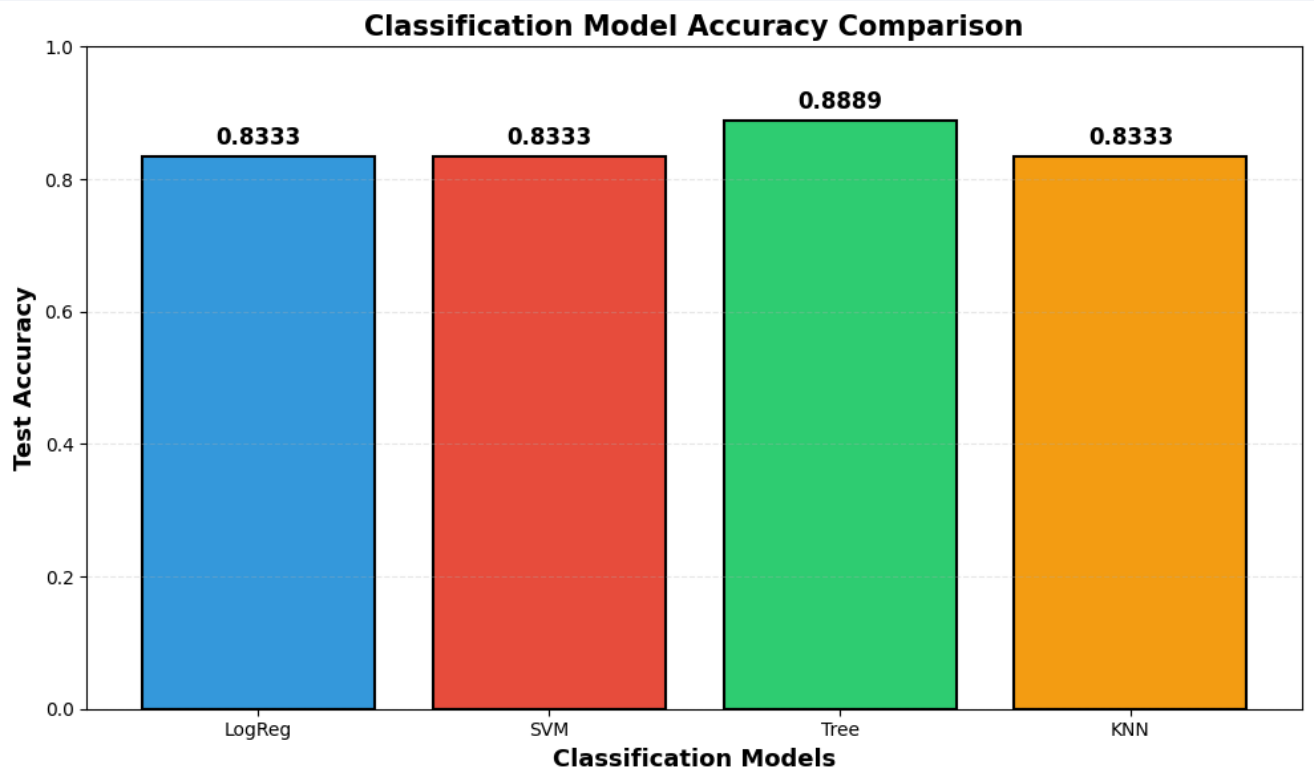
Section 5

Predictive Analysis (Classification)



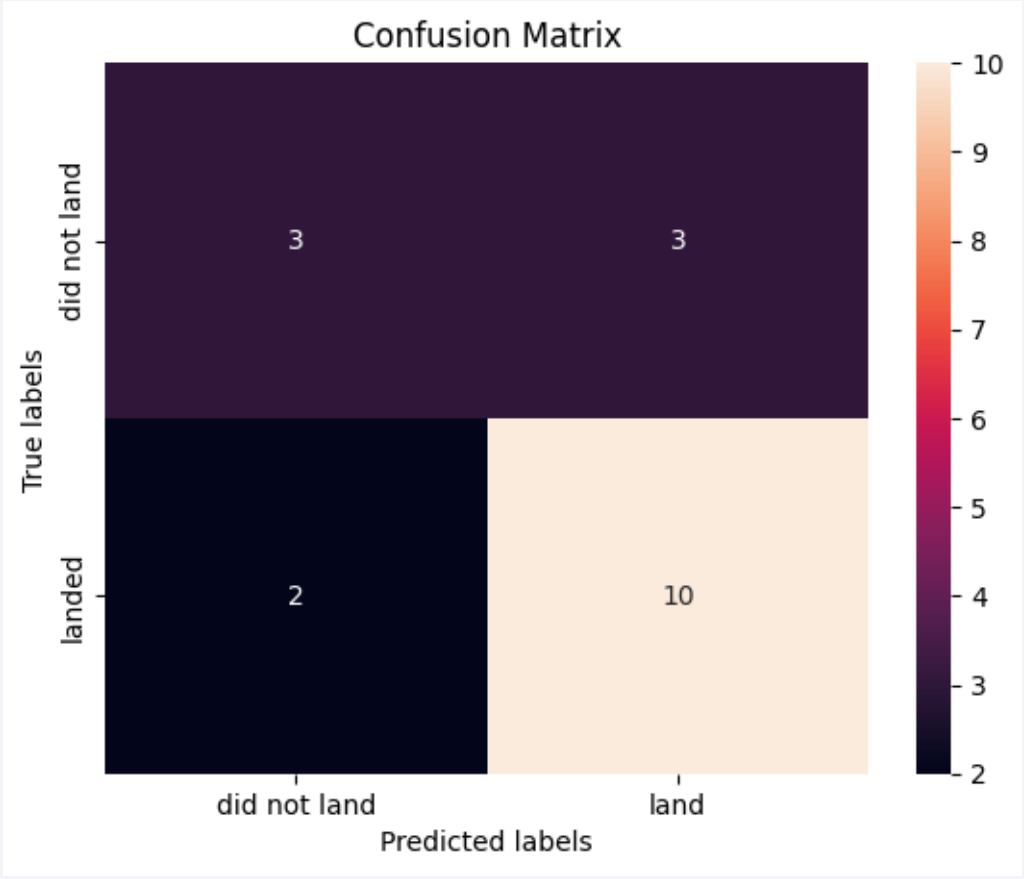
Classification Accuracy

- HIGHEST CLASSIFICATION ACCURACY
  - Model: Tree
  - Test Accuracy: 0.8889 (88.89%)
  - Validation Accuracy: 0.8750



Confusion Matrix

- **Detailed Breakdown**
  - **True Negatives (TN)**: Correctly predicted "did not land"
  - **False Positives (FP)**: Wrongly predicted "landed" (Type I Error)
  - **False Negatives (FN)**: Wrongly predicted "did not land" (Type II Error)
  - **True Positives (TP)**: Correctly predicted "landed"
- **Key Insights**
  - Total correct vs incorrect predictions
  - Percentage accuracy
  - Identifies the main error type (false positives or false negatives)



Conclusions

- **All 4 models achieved 83.33% accuracy** (15/18 correct predictions)
  - Logistic Regression
  - SVM
  - Decision Tree (Best - 88.75% validation)
  - KNN
- **Key Findings:**
  - ✅ **High accuracy** predicting first stage landings
  - ⚠️ **Main error**: False positives (~3 cases) - models predict "landed" when it didn't
  - ✅ **Consistent performance** across all models
- **Business Impact:**
  - **Competitors can estimate SpaceX launch costs with 83% confidence**, enabling competitive bidding.
- **Recommendation:**
  - Use **Decision Tree** - highest validation score (88.75%)
- **Bottom Line:**
  - Machine learning successfully predicts Falcon 9 landing outcomes, making launch cost estimation feasible for SpaceX competitors.

Appendix

- Relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets created during this project can be found at my GitHub repository:
  - https://github.com/antazo/ibm-data-science-ai/tree/main/data-science

Thank you!

